

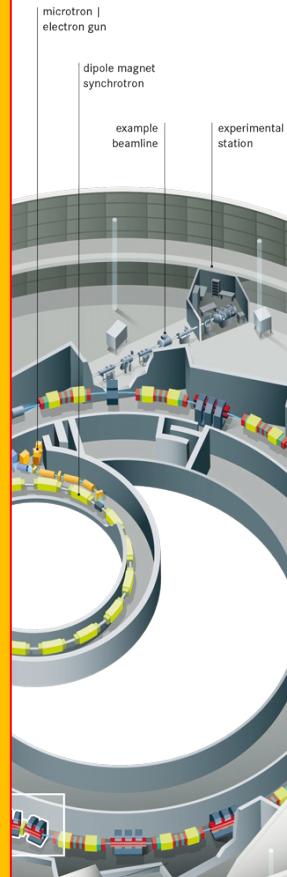
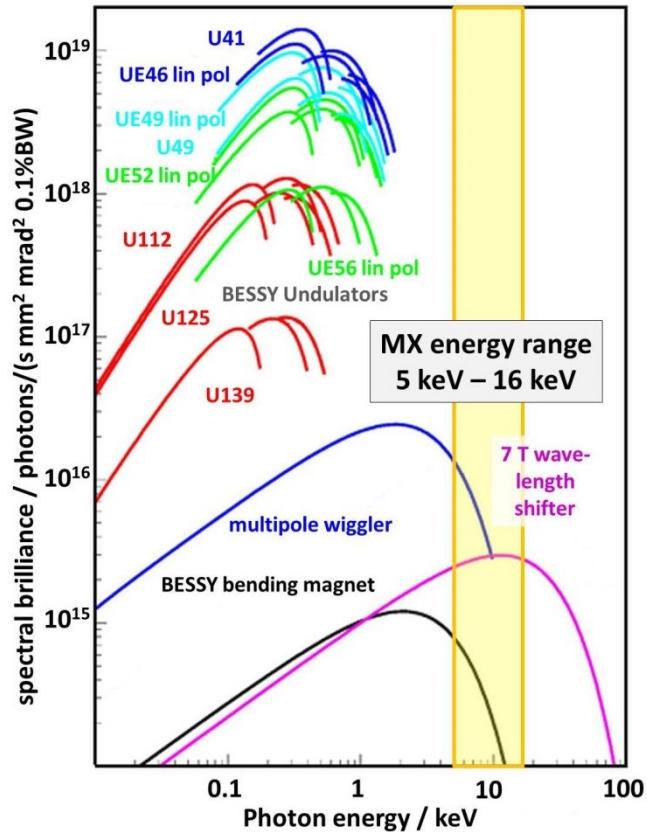
Status of the MX Beamlines at BESSY II

Michael Hellmig,
on behalf of the HZB-MX group

MXCuBE Pre-Meeting, 21.05.2026, on-line



Synchrotron sources at BESSY II



7 Tesla wavelength shifter and MX Beamlines

BESSY II ring parameters:

Electron Energy:	1.7 GeV
Electron Current:	300 mA
Circumference:	240 m
Straight sections:	16
Experimental stations:	~45

main building
„Helmholtz-Zentrum Berlin“

MX experimental floor at BESSY II

BL14.1 (5.5-15.5keV)

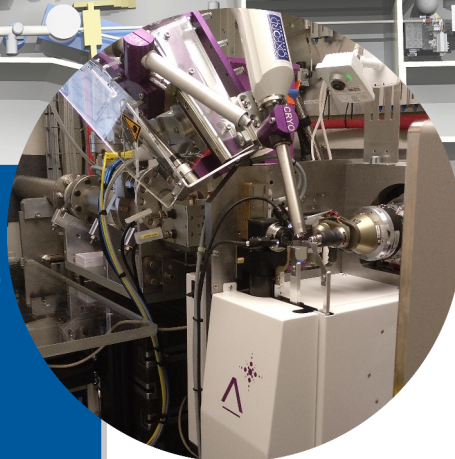
- MD2 with MK3
- Pilatus3 6M 100 Hz
- CATS: 144 UNIPUCK samples



- standard user operation schedule:
24/5 (Tuesday to Saturday)

BL14.3 (13.8 keV)

- MD2S with MK3
- Pilatus3 6M 25 Hz
- HClab & REX nozzle changer



BL 14.2 (5.5-15.5 keV)

- Nanodiff goniometer
- Pilatus3 2M
- ISARA2: 464 UNIPUCK samples

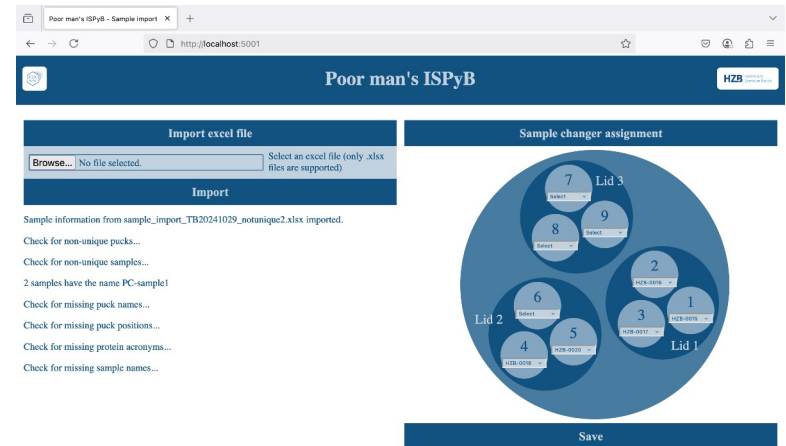


MXCuBE beamline setup (production)

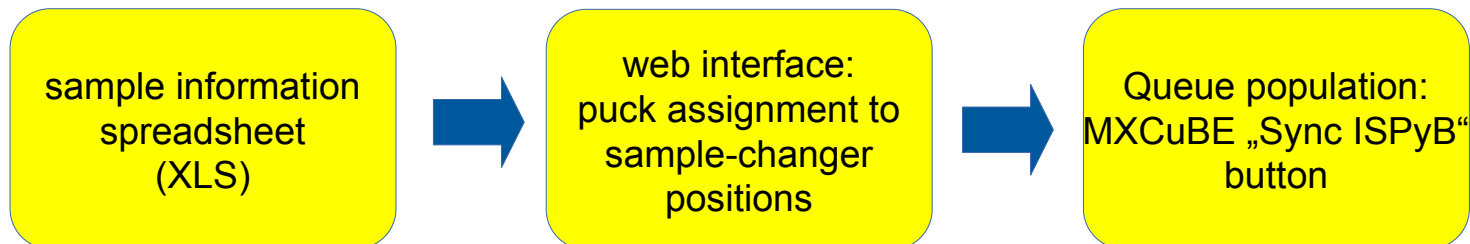
- Qt4 frontend & HWR 2.2 Backend
control systems: Tango, Exporter, EPICS, SPEC
- in-house LIMS solution: “PM-ISPBYB”
 - sample-metadata handling/database
- data processing
 - online auto-processing: XDSApp & Qt results GUI
 - F2X pipeline: FragMAXApp
- beamline IT infrastructure
 - storage: 80 TB beamline + 100 TB F2X (Huawei Dorado)
 - compute:
 - HPE DL580G10 (96C Intel Xeon Platinum 8268, 768 GB RAM)
 - HPE DL380G12 (64C Xeon 6745P, 512 GB RAM)
 - Apple Mac Studio Ultra (24C M2, 128 GB RAM)
- cybersecurity
 - individual network segment per beamline
 - beamline-network access via jumphost
 - remote access: NoMachine with 2FA

- MXCuBE sample metadata import

- implementation by Melanie Oelker
- Python3/SQLite backend
- HTML5/CSS3 frontend
- XLS data-import script
- „automatic“ queue population
- integrated into user's standard workflow



Workflow:



- mxcubeweb/mxcubeqt/mxcubecore:
Github develop branch, 04/2025
- development plan:
basic Qt-based implementation first, switch to web-version later
- BL14.3 prototype setup:
Arinax MD2S
Dectris Pilatus3 6M
manual mounting
- current implementation status:
 - MD2 sample-video integration ✓
 - MD2 motors/components (✓)
 - detector distance (Aerotech motion controller) (✓)
 - EPICS control-system integration: shutters, machine info: WIP

MXCuBE development: plans for the next six months

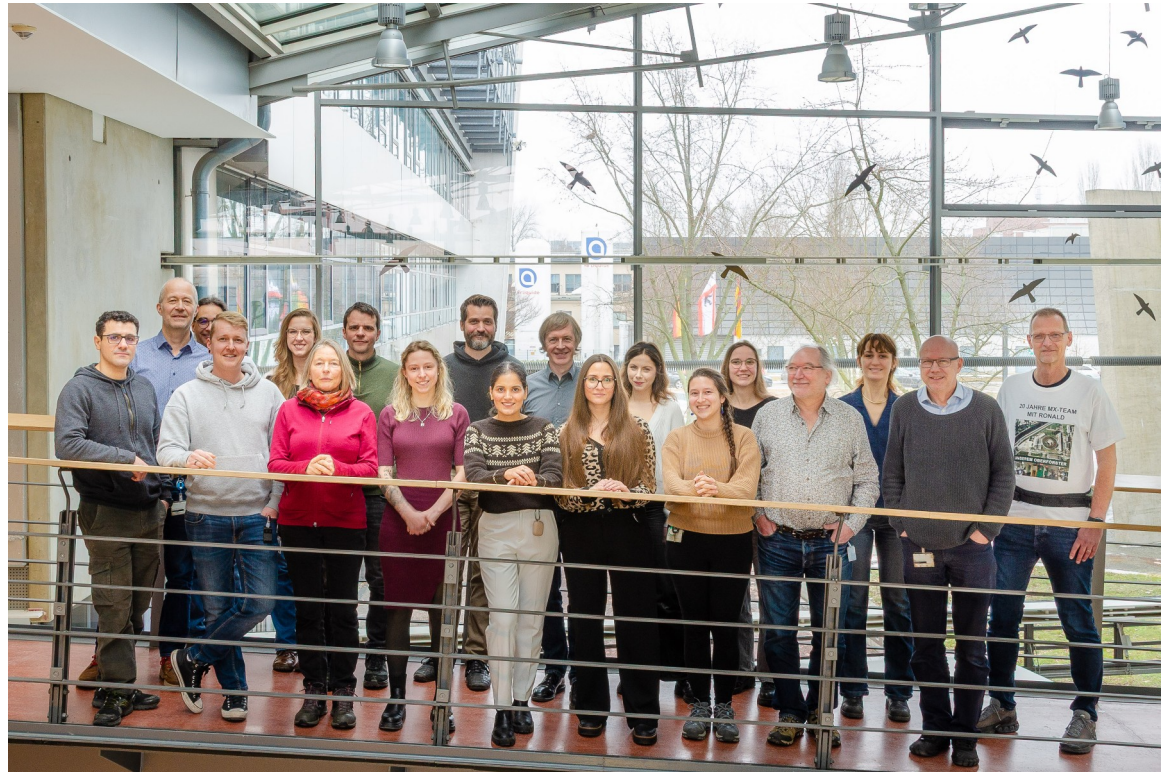
- update development code to current develop HEAD
- complete basic Qt implementation at BL14.3:
 - shutterless data collection
 - EPICS connection to BESSY accelerator/beamline status
 - SPEC beamline control (energy change & beam optimization)
- clone & customize BL14.3 implementation to BL14.1
 - automatic sample mounting: Irelec CATS
- migrate BL14.1/BL14.3 Qt-based implementation to web interface

Acknowledgements

HZB-MX beamline team:

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Markus Wahl



The MXCuBE collaboration